

- Q1.** What is a relation?
 (A) A function with one output
 (B) A set of ordered pairs
 (C) A graph with one line
 (D) A table with one column
 (E) A mapping with one input
- Q2.** How can you determine if a relation is a function using a graph?
 (A) Using the horizontal line test
 (B) Using the vertical line test
 (C) Using the diagonal line test
 (D) Using the slope formula
 (E) Using the equation $y = mx + b$
- Q3.** What is the domain of the function $f(x) = 1 - \frac{1}{x}$?
 (A) All real numbers
 (B) All integers
 (C) All rational numbers except $x = 0$
 (D) All irrational numbers
 (E) The set $\{0\}$
- Q4.** What is the range of the function $g(x) = \sqrt{x}$?
 (A) All real numbers
 (B) All non-negative real numbers
 (C) All non-positive real numbers
 (D) The set $\{0\}$
 (E) The empty set
- Q5.** Evaluate $f(-2)$ for the function $f(x) = x^2 - 3$
 (A) -1
 (B) 1
 (C) 5
 (D) 7
 (E) 11
- Q6.** What is the notation for a function that takes an input x and gives an output y ?
 (A) $y = f(x)$
 (B) $x = f(y)$
 (C) $f(x) = y$
 (D) $f(y) = x$
 (E) $y = g(x)$
- Q7.** Is the relation $\{(1, 2), (2, 3), (3, 2)\}$ a function?
 (A) Yes
 (B) No
 (C) Maybe
 (D) Always
 (E) Never
- Q8.** What type of test can be used to determine if a relation is a function?
 (A) Horizontal line test
 (B) Vertical line test
 (C) Diagonal line test
 (D) Slope test
 (E) Function test

Open Ended Questions (FRQ)

- Q9.** Graph the function $f(x) = 1 - \frac{1}{x}$ and identify its domain and range.

- Q10.** Evaluate the function $g(x) = x^2 - 2x + 1$ for $x = -1, 0, 1$ and write your answers in a table.

- Provide a clear and concise question
- Make sure the question is relevant to the topic
- Use proper notation and formatting